

Multiple Choice (10 marks)

1. Find the inverse Laplace transform $L^{-1}[1 / \{s(s^2 + 1)\}]$, using the convolution.
 - (a) $\sin t - t$
 - (b) $1 - \cos t$
 - (c) $t \sin t$
 - (d) $e^t - 1$
 - (e) $1 - e^{-t}$
2. Which of the following is not a sinusoidal oscillator?
 - (a) Hartley oscillator
 - (b) Colpitts oscillator
 - (c) Clapp oscillator
 - (d) Quartz crystal oscillator
 - (e) LC oscillator
3. Lowest critical frequency is due to zero and it may be present at the origin or nearer to the origin, then the type of network is
 - (a) RL Circuit.
 - (b) RC circuit.
 - (c) CL circuit.
 - (d) CR circuit.
 - (e) RCL circuit.
4. Find the Laplace transform of $f(t) = t^n$, where n is a positive integer.
 - (a) n^n / s^n
 - (b) $n! / s^{n+1}$
 - (c) $n! / s^{n+2}$
 - (d) $(n+1)! / s^n$
 - (e) $n! / s^n$
5. A single phase full bridge inverter can operated in load commutation mode in case load consist of
 - (a) Purely resistive load.
 - (b) RL.

- (c) RLC underdamped with a parallel connection.
- (d) RLC critically damped.
- (e) RLC underdamped with a series connection.

True / False (10 marks)

Answer True or False

6. True or False: In ferromagnetic materials, the magnetisation and applied field are related linearly.
7. True or False: The attenuation per meter along the wave guide for a wavelength of 2 meters is 545 dB/meter.
8. True or False: In a phase-shift keying system with imperfect synchronization, a phase shift of 24° results in $P_e(\text{PSK}) = P_e(\text{DPSK}) = 10^{-5}$.
9. True or False: The solid angle subtended by an area of 2400 cm^2 on a sphere with a diameter of 1.2 m is $1/2$ steradians.
10. True or False: The initial current $i(0^+)$ can be expressed as $E / (v(0^+) * R)$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the change in enthalpy and entropy for a real gas along an isothermal path between pressures P_1 and P_2 , using the provided equation for v and the constants a and b .
12. Discuss the back and front pitches of a 4-pole machine with lap and wave windings, considering the implications of having 36 and 37 winding elements, respectively.

End of Paper